

Which Error Can the District Afford?

Unit 3 · Topic 3.8 · TODAY'S PROBLEM

A district tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, where p is the late-alert proportion. Both plans use $n = 200$ and have the table's operating probabilities. Safety prioritizes avoiding retention when $p = 0.10$; Budget prioritizes avoiding replacement when $p = 0.05$. No common cost scale exists.

Dev: "Use A. Smaller α lowers both error risks."

Lin: "Use B. Its higher power has no cost when safety matters."

Test plans ($n = 200$)

Plan	α	Power at $p = 0.10$
A	0.010	0.748
B	0.050	0.877

FIRST TAKE (5 min, on your sheet)

Which plan would you try first? Name the office you prioritized and one competing cost.

- DOK 1** (a) Describe a Type I error and a Type II error in this context. Identify what $P(\text{reject } H_0 \mid p = 0.10)$ represents.
- DOK 2** (b) For each plan, calculate β . Then give the expected number of Type I errors per 1,000 repeated tests when $p = 0.05$ and Type II errors per 1,000 repeated tests when $p = 0.10$.
- DOK 3** (c) Adjudicate both claims. Explain why no plan is universally best; recommend one for a stated office using both error probabilities and their per-1,000 consequences. Name information needed for a district-wide choice and a change that improves power at fixed α .

Video + follow-along: u6_lesson7_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

